

Curriculum Vitae

PERSONAL INFORMATION

Samir Sadok

Postdoctoral Researcher, Inria Grenoble

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GitHub: <https://github.com/samsad35>

Google Scholar: [link](#)

EDUCATION

- 11/2020–03/2024 **Ph.D. in Signal, Image, and Vision**
CentraleSupélec, IETR (UMR CNRS 6164), France
Thesis: [Audiovisual Speech Representation Learning for Emotion Recognition](#).
Summary: Addressing label scarcity and emotional ambiguity in audiovisual speech emotion recognition through unsupervised and self-supervised deep generative models. Focus on representation learning, disentanglement, and model interpretability.
Advisors: Renaud Séguier, Simon Leglaive
Keywords: Deep representation learning, generative modeling, audiovisual speech processing, emotion recognition.
- 09/2018–08/2020 **M.Sc. in Signal, Image, Systems, and Control (SISEA)**
Université de Rennes 1, ISTIC – Co-accredited with CentraleSupélec
Focus: Theoretical and algorithmic foundations for information processing and decision support.

PROFESSIONAL EXPERIENCE

- 05/2024–2026 **Postdoctoral Researcher at INRIA, Deep Information Geometry**
Inria, Université Grenoble Alpes, CNRS, LJK
Team: RobotLearn.
Studying theoretical and practical links between (Dynamical)VAEs, diffusion models, and normalizing flows. Developing a statistical manifold framework to analyze their training dynamics and improve adaptation efficiency in real-world applications.
- February 2026 **SoRAIM Winter School, Autrans**
Winter School on Social Robotics, Artificial Intelligence, and Multimedia.
- October 2025 **Lab Visit, Spoken language model**
Université de Montréal
Team of Mirco Ravanelli.
- February 2025 **Multi-resolution Flow Matching**
Supervising a master's student on a research project exploring Flow Matching models for multi-resolution generation.
- 06/2022–07/2022 **16th Summer School of Peyresq, Gretschi**
Methodological and societal issues of deep learning.
- 03/2020–08/2020 **Junior Researcher, Orange Labs**
Deep learning for emotion recognition in speech signals.
- 05/2019–08/2019 **Intern, LTSI, University of Rennes 1**
Project: Signal processing for respiratory anomaly detection.

PEDAGOGICAL EXPERIENCE

- 2022-2023 **Teaching - Deep multimodal representation learning** ([Link](#))
Organized within the "Le numérique au service du facteur humain" module for second-year CentraleSupélec students. Covered deep learning-based data fusion, multimodal generative models, and guided students in reading and implementing research papers using PyTorch.
- Spring 2023 **Attention mechanism**
Delivered as part of the same module, this course provided a comprehensive overview of attention mechanisms—from basic seq2seq models to advanced Transformer topics—culminating in a hands-on Transformer encoder implementation.
- 03/2022–05/2022 **Project supervision: Decoding non-verbal behavior**
Guided four student groups in the "Understanding Humans Through Technology" module of CentraleSupélec. Projects involved developing ML/DL approaches for emotion analysis using a common multimodal database, emphasizing literature review, teamwork, and the full workflow from data preparation to model validation.
- 09/2021–01/2022 **Teaching - Practicum supervision**
Supervised practical sessions in digital electronics at the University of Rennes for second-year undergraduates, focusing on FPGA exercises such as implementing a basic calculator.

PUBLICATIONS

My research focuses on deep generative models for audiovisual speech processing, with an emphasis on representation learning, interpretability, and controllability for tasks such as speech generation ([scholar link](#)).

Preprint

1. **Sadok, S.**, Girin, L., & Alameda-Pineda, X. (2026). The Equalizer: Introducing Shape-Gain Decomposition in Neural Audio Codecs. *arXiv*.

International journal

1. **Sadok, S.**, Leglaive, S., & Séguier, R. (2025). A vector quantized masked autoencoder for audiovisual speech emotion recognition. *Computer Vision and Image Understanding*, 104362.
2. **Sadok, S.**, Leglaive, S., Girin, L., Alameda-Pineda, X., & Séguier, R. (2024). A multimodal dynamical variational autoencoder for audiovisual speech representation learning. *Neural Networks*, 106120.
3. **Sadok, S.**, Leglaive, S., Girin, L., Alameda-Pineda, X., & Séguier, R. (2023). Learning and controlling the source-filter representation of speech with a variational autoencoder. *Speech Communication*, 148, 53-65.

International conference

1. **Samir, S.**, Lathuilière, S., & Alameda-Pineda, X. (2026). Residual Tokens Enhance Masked Autoencoders For Speech Modeling. *ICASSP, 2026*.
2. **Sadok, S.**, Hauret, J., Bavu, E. (2025). Bringing Interpretability to Neural Audio Codecs. *INTERSPEECH, 2025*.
3. **Sadok, S.**, Leglaive, S., Girin, L., Richard, G. & Alameda-Pineda, X. (2024). AnCoGen: Analysis, Control and Generation of Speech with a Masked Autoencoder. *ICASSP, 2025*.
4. Belaref, A., **Sadok, S.**, Ibrahim, K., Noumir, Z., & Séguier, R. (2024). Can AI Decode the Circumplex Model of Affect? A Data-Driven Study. *ICPR 2024 workshop*.
5. **Sadok, S.**, Leglaive, S., & Séguier, R. (2023, June). A vector quantized masked autoencoder for speech emotion recognition. In *IEEE ICASSP 2023 Workshop on Self-Supervision in Audio, Speech and Beyond (SASB)*.

TALKS & POSTER SESSION

I have presented my research at various academic events, focusing on deep generative models for speech and multimodal processing, with applications in representation learning and interpretability.

October 2025	ACM SIGMM EU Chapter Kick-off (Link) Dublin, Ireland.
November 2024	GdR IASIS study day on Audio Synthesis (Link) IRCAM, Paris.
May 2024	Talk about analyzing, controlling and generating speech Gibsa-Lab, Grenoble, France.
November 2023	Workshop on Audio Signal Processing (Link) IRCAM, Paris.
December 2023	Extracting interpretable knowledge for the study of spoken communication (Link) Université d'Avignon.
April 12, 2022	Talk at 16ème Congrès Français d'Acoustique (Link) Universitaire Saint Charles, Aix-Marseille Université.
April 01, 2022	Learning of Speech and Language Representations (Link) GDR TAL, Grenoble, France.

SKILLS

Machine Learning & Deep Learning: Generative models (VAEs, GANs, Flow Matching), Representation learning, Self-supervised learning, Multimodal learning.

Programming & Frameworks: Python, PyTorch, NumPy, Scikit-learn, TensorFlow, OpenCV.

Signal & Speech Processing: Multimodal fusion, Audiovisual speech processing, Speech enhancement, Speech synthesis, Emotion recognition.

Mathematical & Theoretical Foundations: Optimization, Probabilistic modeling, Information theory, Riemannian geometry.

Scientific Writing & Communication: Paper writing, Conference presentations, Research collaboration.

LANGUAGES

French (native)

Kabyle (native)

English (very good written and spoken level)

February 17, 2026